

Geometric problems and the methods of solving them

Four standard methods, and a way of choosing between them before you have drawn anything.

LESSON 2

1 × 40 MIN

GEOMETRY 10, §2

IGCSE E4.2

LESSON CLOCK

6'

14'

16'

6'

3'

Four methods	6 min
Choosing between them	14 min
One problem, three methods	16 min
Setting out	6 min
Homework	3 min

LEARNING OBJECTIVES

- Name the synthetic, coordinate, vector and trigonometric methods.
- Choose a method from the wording of a problem.
- Draw a clear labelled diagram before starting.
- Present a solution with reasons at every line.

TEXTBOOK

National	pp. 9–14
Cambridge	Core & Extended, pp. 249–256
Workbook	Method sheet L2

01

Explanation

Four methods

METHOD	TOOLS	REACH FOR IT WHEN
synthetic	congruence, similarity, circle theorems	the problem is about equal or parallel things
coordinate	axes, distance, gradient, equations	perpendicularity, midpoints, or a right angle is convenient
vector	position vectors, scalar product	collinearity, ratios along a line, angles in space
trigonometric	sine and cosine rules, areas	lengths and angles are mixed in the same problem

THE CHOOSING RULE

Look at what the problem **gives**, not at what it asks. Given angles and lengths — trigonometry. Given a rectangle or a right angle — coordinates. Given ratios along segments — vectors. Given equal sides or parallels — synthetic.

One problem, three ways

Problem. $ABCD$ is a square. M is the midpoint of BC . Prove AM and DM are not perpendicular, and find the angle $\angle AMD$.

METHOD	FIRST LINE
coordinate	$A(0,0), B(2,0), C(2,2), D(0,2), M(2,1)$
vector	$AM = 2i + j, DM = 2i - j$
trigonometric	$AM = DM = \sqrt{5}, AD = 2$, then the cosine rule

$$\cos \angle AMD = \frac{5 + 5 - 4}{2 \cdot 5} = 0.6 \Rightarrow \angle AMD \approx 53.1^\circ$$

All three agree. The coordinate method is shortest here because the square hands you the axes.

Setting out

FIVE HEADINGS, EVERY TIME

Given · Required · Diagram · Solution, with a reason on every line · **Answer**, in a sentence with units.

DRAW IT BIG

A diagram smaller than a matchbox hides the very relationship you are looking for. Half a page, labelled, with the given data marked on it — that is where most solutions are actually found.

02 Worked examples

EXAMPLE 1 Which method: “In triangle ABC , $a = 7$, $b = 9$, $\angle C = 40^\circ$. Find c .”

- ① Two sides and the included angle.
Lengths mixed with an angle.

- ② Cosine rule.

ANSWER

Trigonometric method

EXAMPLE 2 Which method: “Prove the diagonals of a rhombus are perpendicular.”

- ① Equal sides given.
Congruent triangles available.

- ② Coordinates also work, but need setting up.

ANSWER

Synthetic (coordinates a close second)

EXAMPLE 3 Which method: “ P divides AB in the ratio $2 : 3$. Show P, Q, R are collinear.”

- ① Ratios along segments, collinearity.
- ② Position vectors handle both directly.

ANSWER

Vector method

03 Interactive model

Solve the square problem on the board by two methods and let the class time each.

Pick the method

Two sides and the included angle, find the third side:

Prove the midpoints of a quadrilateral form a parallelogram:

Find the perpendicular distance from a point to a line:

Prove base angles of an isosceles triangle are equal:

synthetic

coordinate

vector

trigonometric

The first thing to do with any geometry problem:

write the formula

draw a large labelled diagram

guess the answer

measure

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Synthetic method	Sintetik usul	Синтетический метод
Coordinate method	Koordinatalar usuli	Координатный метод
Vector method	Vektor usuli	Векторный метод
Trigonometric method	Trigonometrik usul	Тригонометрический метод
Auxiliary line	Yordamchi chiziq	Вспомогательная линия
Construction	Yasash	Построение
Analysis of a problem	Masalani tahlil qilish	Анализ задачи
Given and required	Berilgan va topilishi kerak	Дано и найти
Diagram	Chizma	Чертёж

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The coordinate method is best when the figure has:

equal sides

right angles

circles only

no data

Collinearity is most directly proved with:

vectors

the sine rule

congruence

measurement

A reason must be written:

at the end

on every line

only in proofs

never

An auxiliary line is:

given

added by you to help

always wrong

a mistake

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the four methods

ANSWER synthetic, coordinate, vector, trigonometric

2. Best method for the cosine rule situation

ANSWER trigonometric

3. Best method for collinearity

ANSWER vector

4. Best method for “prove the diagonals bisect”

ANSWER synthetic or vector

5. What are the five headings of a solution?

ANSWER given, required, diagram, solution, answer

6. How large should a diagram be?

ANSWER about half a page

7. What is an auxiliary line?

ANSWER A line you add to create a usable figure

Medium · the standard question

7 PROBLEMS

1. Square side 2, M midpoint of BC . Find AM

ANSWER $\sqrt{5}$

2. Same square; find $\angle AMD$

ANSWER $\approx 53.1^\circ$

3. Choose a method: find the angle between two space diagonals of a cube

ANSWER **vector**

4. Choose a method: prove a midline is half the base

ANSWER **vector or synthetic**

5. Choose a method: a triangle with sides 5, 6, 7 — find its largest angle

ANSWER **trigonometric**

6. Set up coordinates for a rectangle 6×4

ANSWER $(0,0), (6,0), (6,4), (0,4)$

7. Why is a big diagram worth the time?

ANSWER **It reveals the relationship you must use**

Hard · stretch and reason

7 PROBLEMS

1. Solve the square-and-midpoint problem by all three methods and compare lengths of solution

ANSWER Coordinate is shortest

2. Prove the medians of a triangle are concurrent, by vectors

ANSWER Show $\frac{a+b+c}{3}$ lies on each median

3. Prove the midpoints of any quadrilateral form a parallelogram

ANSWER Each midline is $\frac{1}{2}$ of a diagonal

4. A cube of side 1: find the angle between a face diagonal and a space diagonal

ANSWER $\approx 35.26^\circ$

5. When would the coordinate method be a bad choice?

ANSWER When the figure has no right angles or convenient symmetry

6. Give a problem where two methods take about the same time

ANSWER Diagonals of a rhombus

7. Explain why the vector method extends to space but the plane coordinate method must be extended first

ANSWER Vectors already carry three components

07 Homework

Homework — 4 tasks

Task 2 needs the diagram, not only the answer.

1. For five problems from the exercise, write only the method you would choose and one sentence of why.
2. Solve the square-and-midpoint problem by the coordinate method, setting out all five headings.
3. Solve it again by the cosine rule and check the two answers agree.

4. Find a problem in the textbook where the vector method is clearly best, and say why.